THE AMINO ACID AND FATTY ACID PROFILE OF CALIFORNIA PIGEONS (*COLUMBA LIVIA*) MEAT

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Introduction

Interest in non-traditional poultry species is increasing due to their potential as high-quality protein sources with favourable lipid composition. California pigeons are a meat-type breed characterized by rapid growth and high carcass yield. However, data on the detailed amino acid and fatty acid composition of their meat remain limited. Understanding these parameters is essential for evaluating nutritional value and functional

Objective

To evaluate the amino acid and fatty acid composition of breast and thigh muscles in male and female California pigeons, with emphasis on nutritional quality and muscle-related differences.

Materials and Methods

Animals: 20 males, 20 females, 3 months of age

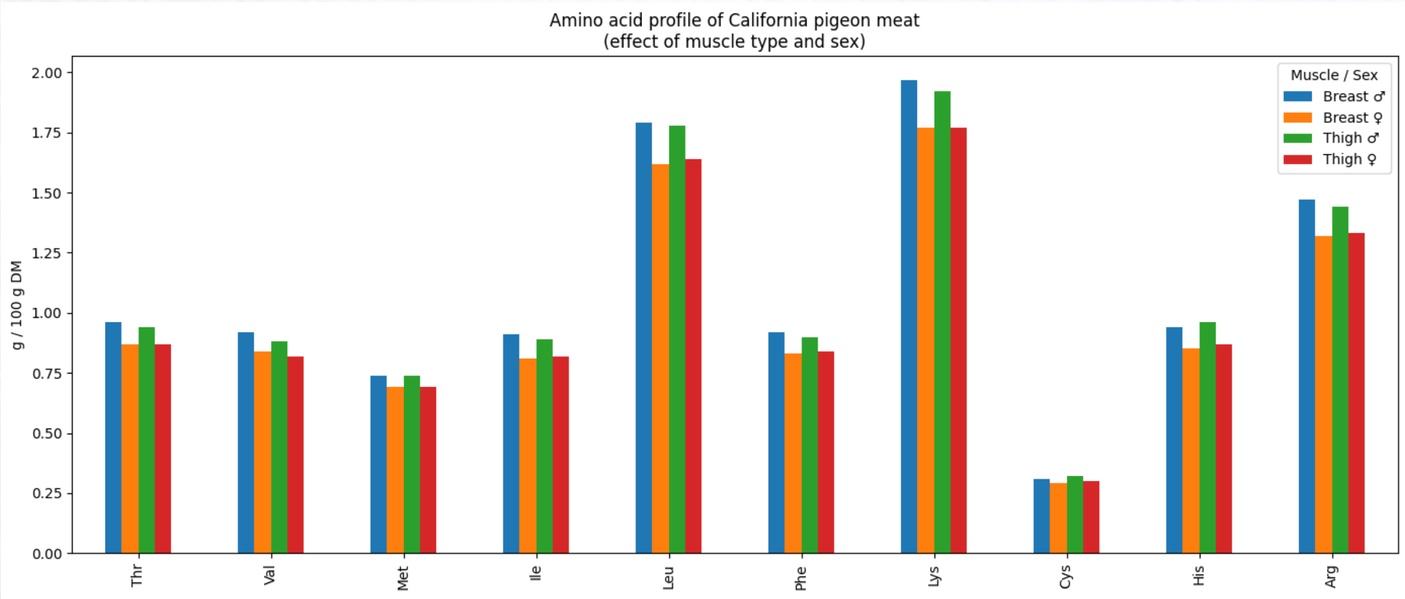
Muscles analysed: breast and thigh

Amino acids: Automatic Amino Acid Analyzer (AA

Fatty acids: Gas chromatography of FAME

Statistics: ANOVA, Scheffe test, $p \leq 0.05$

Results

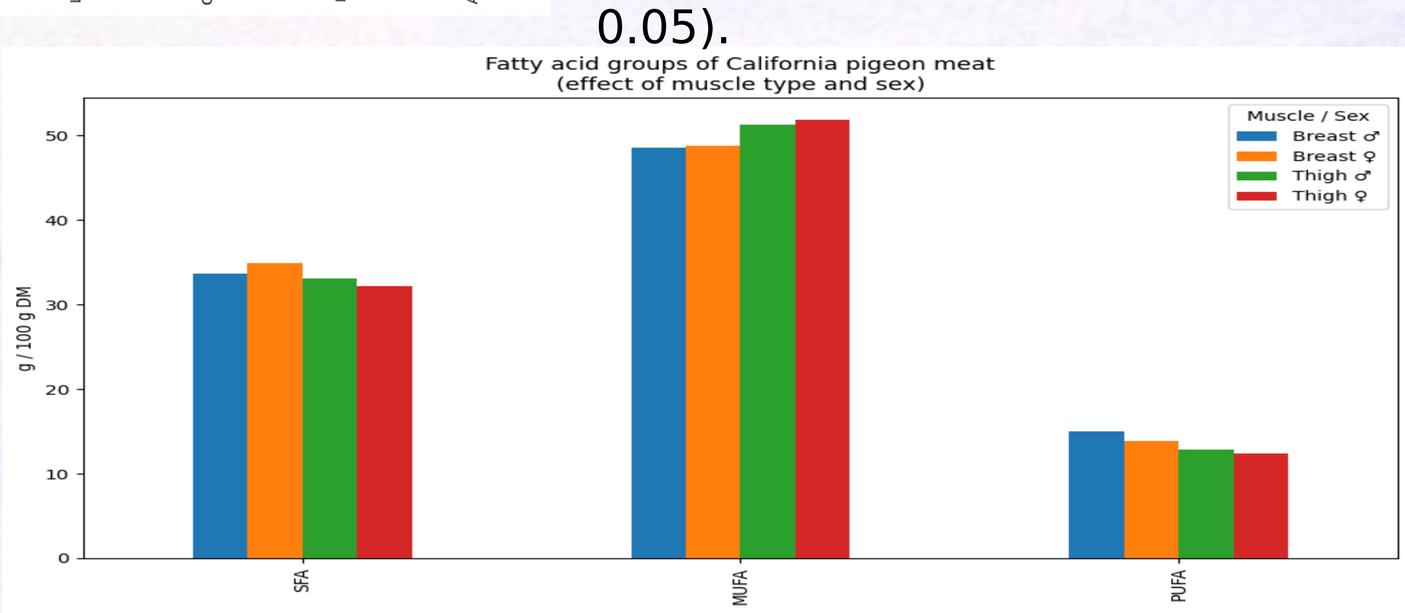


Breast muscle exhibited slightly higher concentrations of essential amino acids, particularly lysine and leucine, compared to thigh muscle, with consistently higher values observed in males.

No significant effects of sex or muscle on AA composition ($p > 0.05$).

Monounsaturated fatty acids represented the predominant lipid fraction in both muscles and sexes, with higher MUFA proportions observed in thigh muscle.

FA composition was not significantly affected by sex or muscle ($p > 0.05$).



Conclusion

California pigeon meat exhibits a high-quality amino acid profile with a substantial proportion of essential amino acids and a nutritionally favourable lipid composition dominated by monounsaturated fatty acids. Although differences between sexes and muscle types were not statistically significant, consistent biological trends were observed. These results confirm the potential of California pigeon meat as a premium, health-oriented alternative to conventional poultry meat.

Acknowledgments:

This research was funded by the grant APVV-23-0077 “Use of venison-local sustainable source of nutrients on the Slovak market “